# iLearn Practice Exam

11BR · Chapter / Topic 5

## Cambridge AS Level Computer Science 9618

Paper 1  Theory Fundamentals

# Topic 5: System Software

**Name:** _____

**Class:** _____

**Date:** _____

**Time allowed:** 1 hour 30 minutes

**TOTAL MARKS: 74**

Marks obtained: ..... / 74

**INSTRUCTIONS**

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

**INFORMATION**

· The total mark for this paper is 74.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 8 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, mark your work against the official Cambridge mark scheme (see the source list on the last page).

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A student has purchased a new laptop.

(a) The laptop has systems software.

**(i)** Describe how the Operating System (OS) manages processes in the computer.

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..... [5]

**(ii)** Describe the purpose of utility software in a computer.

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..... [2]

2 A software developer is working in a team writing a program for a client.

**(a)** The developer is writing a new program library to be used by the other team members.

**(i)** Define the term **program library**.

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..... [2]

**(ii)** Explain **two** benefits to the developer of choosing to create a Dynamic Link Library (DLL).

1 .....

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2 .....

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[4]

**(b)** The development team needs to use a translator whilst writing the program for the client.

Identify whether an interpreter or a compiler would be more appropriate at this stage of the program development.

Justify your choice.

Translator .....

Justification .....

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..... [3]

**(c)** The development team uses an Integrated Development Environment (IDE).

Complete the table by describing the typical features found in an IDE.

| Feature | Description |
|---|---|
| Breakpoints | ..... ..... ..... |
| Dynamic syntax checks | ..... ..... ..... |
| Context-sensitive prompts | ..... ..... ..... |
| Single stepping | ..... ..... ..... |

[4]

3 A computer has an Operating System (OS).

Memory management and process management are two OS tasks.

Explain how memory management **and** process management support multi-tasking.

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..... [4]

4 A programmer uses an Integrated Development Environment (IDE) to write a computer program. The IDE has both a compiler and an interpreter as built-in translators.

**(a)** The programmer decides to use the compiler when testing the final program.

Describe the benefits of using the compiler during testing.

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..... [2]

**(b)** IDEs have many features other than built-in translators.

Complete the table by identifying **one other** common IDE feature that can be used for each purpose. Describe how each feature helps the user during program development.

Each feature must be different. Do **not** give translator as one of your features.

| Purpose | IDE feature | Description |
|---|---|---|
| for coding | ..... ..... | ..... ..... ..... ..... |
| for presentation | ..... ..... | ..... ..... ..... ..... |
| for debugging | ..... ..... | ..... ..... ..... ..... |

[6]

**(c)** The programmer uses program libraries when developing the program.

Describe **two** benefits to the programmer of using program libraries.

1 .....

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2 .....

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[2]

5 A student is writing a computer program for a school project.

**(a)** The student uses a computer that comes with an Operating System (OS) and utility software.

**(i)** Identify **two** tasks the OS performs as part of process management.

1 .....

2 .....
[2]

**(ii)** Identify **two** tasks the OS performs as part of memory management.

1 .....

2 .....
[2]

**(iii)** Explain the reasons why defragmentation software improves the performance of a computer.

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..... [3]

**(b)** The student is using an Integrated Development Environment (IDE).

**(i)** Explain how the student can use debugging tools in the IDE.

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..... [4]

**(ii)** The high-level language the student is using can be partially compiled and partially interpreted.

Explain the reasons why some high-level languages are partially compiled and partially interpreted.

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..... [2]

6 A student uses a laptop for school work. The laptop contains system software including an Operating System (OS) and utility software.

**(a)** Hardware management is one key management task that the OS performs.

**(i)** Identify **two** tasks the OS performs as part of hardware management.

1 .....

2 .....
[2]

**(ii)** Identify **two** other key management tasks that the OS performs.

1 .....

2 .....
[2]

**(b)** A disk formatter is an example of utility software.

**(i)** State the purpose of utility software in a computer.

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..... [1]

**(ii)** Describe the purpose of a disk formatter.

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..... [2]

**(c)** The student is writing a computer program in a high-level language.

The student imports routines from program libraries into their program.

**(i)** Define the term program library.

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..... [1]

**(ii)** Describe **one** benefit of the student using a Dynamic Link Library (DLL) instead of a non-Dynamic Link Library.

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..... [2]

7 A programmer uses an Integrated Development Environment (IDE) to develop a program that monitors air quality.

**(a)** Describe the following features of a typical IDE.

Context-sensitive prompts .....

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Single stepping .....

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[4]

**(b)** The program is distributed by downloading the source code and its library files from a web server.

**(i)** Explain the reasons for compressing the files.

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..... [2]

**(ii)** The program files are stored on a new hard disk after they have been downloaded.

Describe the reasons why a hard disk formatter is needed for the new hard disk.

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..... [3]

8 Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

**(a)** State **two** disadvantages of using a compiler compared to an interpreter during program development.

1 .....

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2 .....

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[2]

**(b)** Explain how a programmer benefits from using program libraries during program development.

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..... [3]

(c) Memory management is one key management task performed by an OS.

Give **three** other key management tasks that are performed by an OS.

1 .....

2 .....

3 .....

[3]

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**END OF QUESTION PAPER**

## Source of each question (for marking)

Find the matching official Cambridge mark scheme for the series/paper shown, to mark your answers.

| Q | Past paper | Original question | Marks |
|---|---|---|---|
| 1 | May/June 2023 Paper 12 | Q5 (d) | 7 |
| 2 | May/June 2023 Paper 12 | Q7 (a),(b),(c) | 13 |
| 3 | May/June 2024 Paper 12 | Q6 | 4 |
| 4 | May/June 2024 Paper 12 | Q8 (a),(b),(c) | 10 |
| 5 | May/June 2026 Paper 12 | Q7 (a),(b) | 13 |
| 6 | May/June 2026 Paper 13 | Q5 (a),(b),(c) | 10 |
| 7 | Oct/Nov 2023 Paper 13 | Q5 (a),(b) | 9 |
| 8 | Oct/Nov 2025 Paper 13 | Q8 (a),(b),(d) | 8 |

**Total: 74 marks**